

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 15-02-26
Time: 3 Hrs Max. Marks: 300

15-02-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-2_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

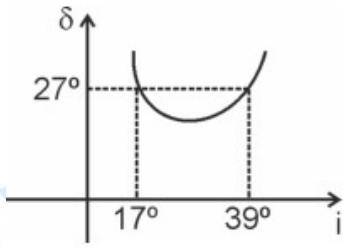
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

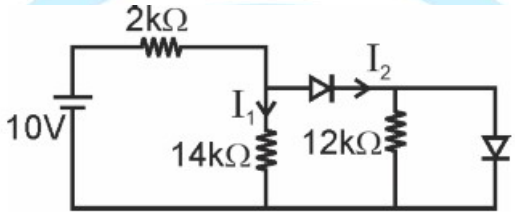
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Deviation vs angle of incidence graph for a prism is shown in the figure, angle of prism will be :-



(A) 20° (B) 29° (C) 27° (D) 25°

2. If both the diodes are ideal then value of I_1 & I_2 will be :-



(A) 0,0 (B) 5 mA, 5 mA (C) 5 mA, 0 (D) 0, 5 mA

3. **Assertion :-** A planet is moving around the sun in an elliptical orbit, then from aphelion point to perihelion point, the speed of planet continuously increases.
Reason :- According to Kepler's second law, the areal velocity of the line joining planet to sun is constant.
(A) Assertion & Reason are true and reason is correct explanation of assertion
(B) Assertion & reason are true & reason is not correct explanation of assertion,
(C) Assertion is true & reason is false.
(D) Assertion is false & reason is true.

4. Velocity of a particle is given as $\vec{v} = x^2\hat{i} + 2y^2\hat{j}$. Then the equation of trajectory if at $t = 0$ particle is at $(4,1)$ is :-

(A) $x + y - xy = 0$ (B) $xy - 2x + 4y = 0$
(C) $x - xy + 4y = 0$ (D) $2xy - 2x + y = 0$

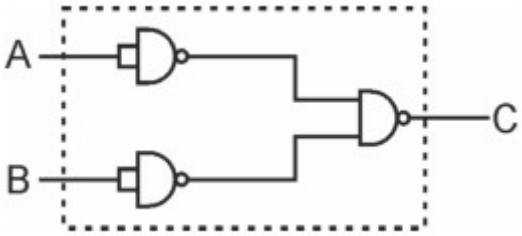
5. $F = At - \dfrac{A}{x + Bm}$; where, $F =$ force, $t =$ time, $x =$ displacement, $m =$ mass, A, B are constants.

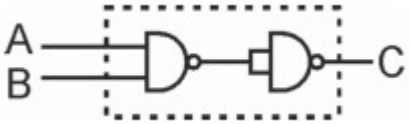
Dimensional formula of $\dfrac{A}{B}$ is:

(A) $[M^2\ T^3]$ (B) $[M^2\ T^{-3}]$ (C) $[MT^{-3}]$ (D) $[M^{-3}\ T]$

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6. The combination of NAND gates are equivalent to



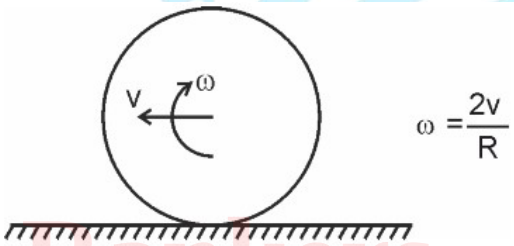


(A) OR gate and AND gate
(B) AND gate and NOT gate
(C) AND gate and OR gate
(D) OR gate and NOT gate

7. There is a bulb of power 100 W which converts 6% of power consumption in electromagnetic wave. The amplitude of electric field at a distance 1 m from the bulb is :

(A) $6\ V/m$ (B) $6\sqrt{10}\ V/m$ (C) $10\ V/m$ (D) $5\ V/m$

8. An object is placed on rough horizontal surface as shown in figure. Initially it has velocity v towards left & angular velocity ω as shown in figure. After sometime, the object will start pure rolling. Determine direction of velocity of centre of mass (left or right), when pure rolling begins. (Radius of object is R)



(A) Right if object is ring, right if object is solid sphere
(B) Right if object is ring, left if object is solid sphere
(C) Left if object is ring, right if object is solid sphere
(D) Left if object is ring, left is object is solid sphere

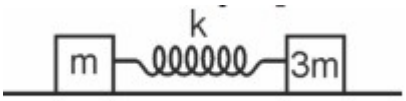
9. Transverse wave travelling along the string from left to right is incident normally on a boundary at $x = 0$. Incident wave is reflected from free end or fixed end. List I contains the equation of incident wave and end from which it is reflected. List II contains the equation of reflected wave.

List-I		List-II	
(I)	$y = A\sin(kx - \omega t)$, free end	p)	$y = -A\cos(kx + \omega t)$
(II)	$y = A\sin(kx - \omega t)$, rigid end	q)	$y = A\sin(kx + \omega t)$
(III)	$y = A\cos(kx - \omega t)$, free end	r)	$y = -A\sin(kx + \omega t)$
(IV)	$y = A\cos(kx - \omega t)$, rigid end	s)	$y = A\cos(kx + \omega t)$

(A) I → p, II → q, III → r, IV → s
(B) I → r, II → q, III → s, IV → p
(C) I → s, II → q, III → s, IV → p
(D) I → s, II → p, III → q, IV → p

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10. Two blocks of mass m & 3 m are placed on smooth horizontal surface, both blocks are connected by ideal spring of spring constant k . The spring is now stretched by & released. The amplitude of oscillation of mass m is :-



(A) $\frac{\ell}{3}$ (B) $\frac{2\ell}{3}$ (C) $\frac{\ell}{4}$ (D) $\frac{3\ell}{4}$

11. A nucleus with mass number 184 initially at rest emits an α-particle. If the Q value of the reaction is 5.5 MeV , the kinetic energy of α-particle is :-

(A) 5 MeV (B) 5.5 MeV (C) 0.12 MeV (D) 5.38 MeV

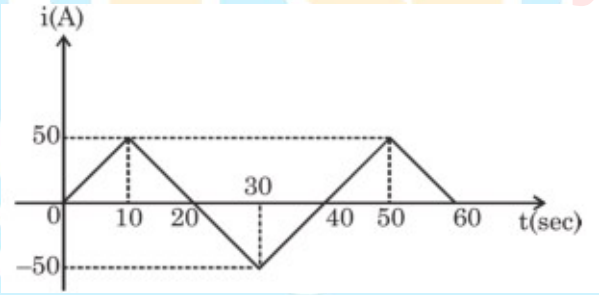
12. A piece of metal weighs 45 gm in air and 25 gm in a liquid of density $1.5\times10^{3}\,\text{Kgm}^{-3}$ is kept at 30°C. When the temperature of the liquid is raised to 40°C, the metal piece weighs 27 gm . The density of liquid at 40°C is $1.25\times10^{3}\ \text{kg}-\text{m}^{-3}$. The coefficient of linear expansion of metal is :-

(A) $1.3\times10^{-3}\ /^{\circ}\text{C}$ (B) $5.2\times10^{-3}\ /^{\circ}\text{C}$ (C) $2.6\times10^{-3}\ /^{\circ}\text{C}$ (D) $0.26\times10^{-3}\ /^{\circ}\text{C}$

13. In a Fraunhoffer diffraction experiment at a single slit using light of wavelength 400 nm , the first minimum is formed at an angle of 30°. Then the direction θ of the first secondary maximum is :-

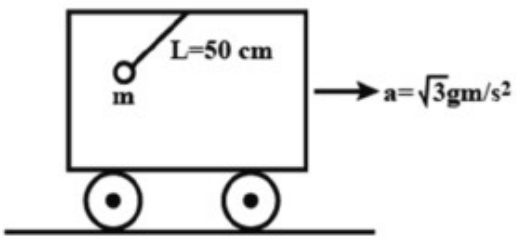
(A) $\tan^{-1}\left(\frac{4}{3}\right)$ (B) 60° (C) $\left(\frac{3}{4}\right)$ (D) $\tan^{-1}\left(\frac{3}{4}\right)$

14. Value of i_{rms} for the following graph for $t=0$ to $t=60$ s is :



(A) $50\sqrt{3}$ A (B) $\frac{50}{\sqrt{3}}$ A (C) 50 A (D) None of these

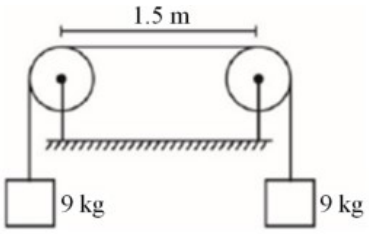
15. A simple pendulum 50 cm long is suspended from the roof of a cart accelerating in the horizontal direction with constant acceleration $\sqrt{3}\text{gm}/\text{s}^2$. The period of small oscillations of the pendulum about its equilibrium position is ($g=\pi^2\ \text{m}/\text{s}^2$)



(A) 1.0 sec (B) $\sqrt{2}$sec (C) 1.53 sec (D) 1.68 sec

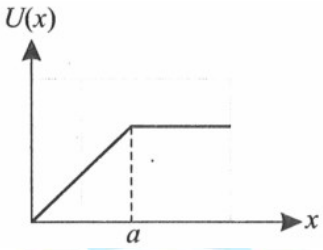
For More Join: @IITJEE_Advanced

16. Consider the following pulley mass system. The frequency of the vibration of the second overtone of the wire between the pulleys is (Given: Mass of the wire between the pulleys is : 24 g)

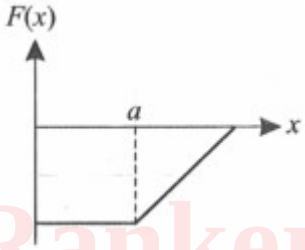


A) 25 Hz B) 115 Hz C) 50 Hz D) 75 Hz

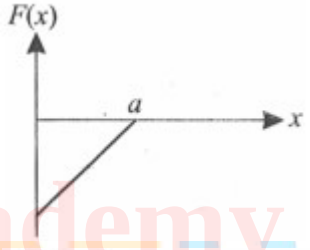
17. The potential energy of a system is represented in the first figure. The force acting on the system will be represented by



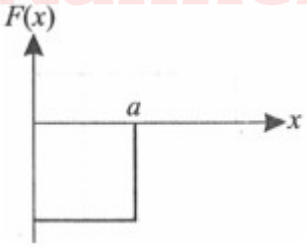
A)



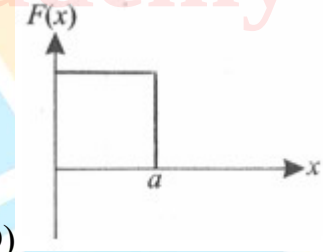
B)



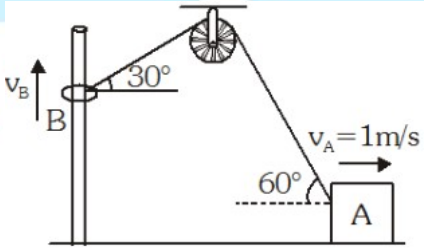
C)



D)



18. Find velocity of ring $B(V_B)$ at the instant shown. The string is taut and inextensible :



(A) $\frac{1}{2}$ m/s (B) $\frac{\sqrt{3}}{4}$ m/s (C) $\frac{1}{4}$ m/s (D) 1 m/s

19. In an electromagnetic wave, the electric field vector and magnetic field vector are given as $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{k}$, respectively. The direction of propagation of electromagnetic wave is along

(A) $-\hat{k}$ (B) $-\hat{j}$ (C) $\hat{j}$ (D) $\hat{k}$

For More Join: @IITJEE_Advanced

20. A paramagnetic samples shows intensity of magnetization of $0.8\ A/m$, when placed in an external magnetic field of strength 0.8 T at a temperature 5 K . When the sample is placed in an external magnetic field of 0.4 T at a temperature of 20 K , the intensity of magnetization is :

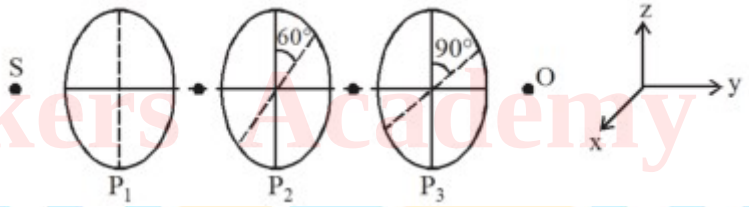
(A) $10^{-1}\ A/m$ (B) $10^{-2}\ A/m$ (C) $10\ A/m$ (D) $10^{-3}\ A/m$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

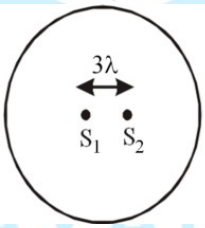
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A charge Q is uniformly distributed over a large plastic sheet. The electric field at a point P close to centre of plate is $10\ V/m$. If the plastic plate is replaced by a copper plate of the same geometrical dimensions and carrying the same charge Q , the electric field at the point P will be (in V/m).

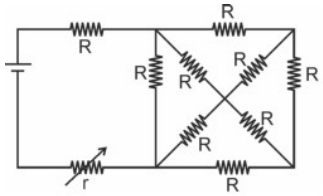
22. As shown in figure, three identical polaroids P_1, P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1. The source S of unpolarised light has an intensity of $256\frac{W}{m^2}$. The intensity of light at point O is $-----\frac{W}{m^2}$.



23. If two coherent sources are placed at a distance 3λ from each other symmetric to the centre of the circle shown in the figure, then number of maxima shown on the screen placed along the circumference is :-



24. In the circuit shown, the resistance r is variable, if heat generation in r is maximum, then the value of r(in Ω) is : (If $R = 15\Omega$)



25. A hydrogen like atom of atomic number Z is in excited state of quantum number 2 n . It can emit a maximum energy photon of 204 eV. If it makes a transition from 2n to the quantum state n then a photon of energy 40.8 eV is emitted. Calculate the atomic number Z.

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Calculate molar mass (in gm/mol) of non-volatile solute, whose 50 gm is dissolving in $250\ gH_2O$, produces a solution having boiling point $110.4^\circ C$. Van't Hoff factor is $4\left(K_b \text{ of } H_2O = 0.52\ K.kg.mol^{-1}\right)$

(A) 40 (B) 50 (C) 30 (D) 60

27. Which of the following f-block element doesn't form MO_2 type oxide

(A) Pr (B) Nd (C) Dy (D) Tm

28. **Assertion :** Treatment of phenol with nitrous acid yields p-benzoquinone monoxime.

Reason : p-nitrosophenol and p-benzoquinone monoxime are tautomers.

(A) Both assertion and reason are true and reason is correct explanation of assertion.

(B) Both assertion and reason are true and reason is not the correct explanation of assertion.

(C) Assertion is true but reason is false.

(D) Assertion is false but reason is true.

29. When two sparingly soluble salts A & B share a common ion, their simultaneous solubility :-

(A) Exceeds their individual solubilities in pure water

(B) Lies between their separate solubilities.

(C) Decrease for more soluble but increase for less soluble salt.

(D) Is less than their individual solubilties in pure water.

30. In 100 ml solution of mohr's salt contains 6.4 mg of ' S '. The molarity of ammonium ion in solution (Aw of $Fe = 56,\ S = 32, O = 16,\ N = 14$)

(A) 4×10^{-3} molar (B) 2×10^{-3} molar

(C) 6×10^{-3} molar (D) 2×10^{-4} molar

31. The correct order of Δ_0 (the octahedral crystal field splitting of d orbitals) value for the following anionic metal complexes is

(A) $[Ir(CN)_6]^{3-} < [Rh(CN)_6]^{3-} < [RhCl_6]^{3-} < [CoCl_6]^{2-}$

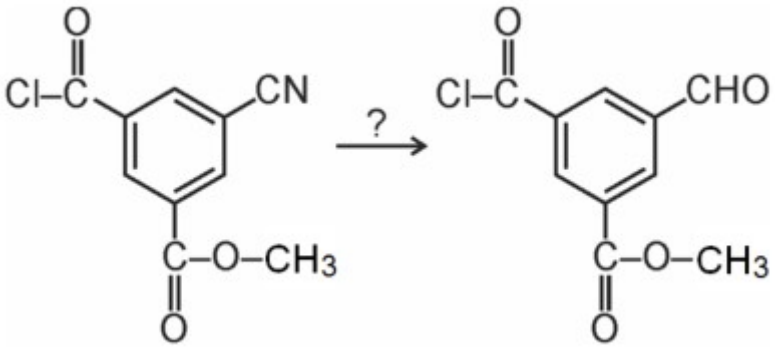
(B) $[CoCl_6]^{3-} < [RhCl_6]^{3-} < [Rh(CN)_6]^{3-} < [Ir(CN)_6]^{3-}$

(C) $[CoCl_6]^{3-} < [Rh(CN)_6]^{3-} < [RhCl_6]^{3-} < [Ir(CN)_6]^{3-}$

(D) $[Ir(CN)_6]^{3-} < [CoCl_6]^{3-} < [Rh(CN)_6]^{3-} < [RhCl_6]^{3-}$

For More Join: @IITJEE_Advanced

32.



Suitable reducing agent is?

(A) $SnCl_2 + HCl$ (followed by H_3O^+)

(B) DIBAL-H/-78 °C

(C) $H_2 / Pd, BaSO_4$

(D) $LiAlH_4$

33. From the given following data :

$A^{2+} + 2e^- \rightarrow A;\ E^\circ = -0.15\ V$

$A^{4+} + 2e^- \rightarrow A^{2+};\ E^\circ = 0.12\ V$

Which of the following statement is correct?

(A) A^{2+} is unstable and disproportionates to A^{4+} and A .

(B) A^{2+} is stable and disproportionation reaction is not spontaneous.

(C) A can't liberate hydrogen from dilute acidic solution

(D) A^{+4} is most stable oxidation state of A .

34. Match List-I with List-II

List-I		**List-II** (No. of lone pair of electrons on the central atom)	
(a)	I_3^-	(i)	0
(b)	IF_5	(ii)	1
(c)	IF_7	(iii)	2
(d)	IF_3	(iv)	3

Choose the most appropriate answer from the options given below :-

(A) a →iv, b → i, c → ii, d → iii

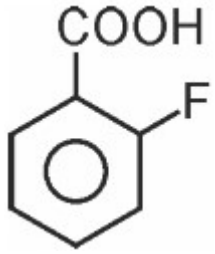
(B) a →iii, b → iv, c → ii, d → i

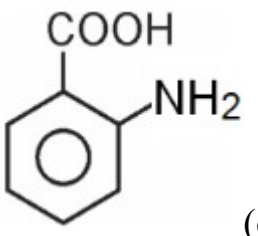
(C) a → iii, b → ii, c → iv, d → i

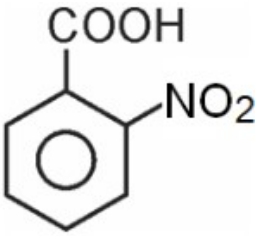
(D) a → iv, b → ii, c → i, d → iii

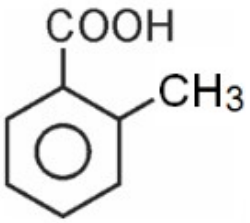
For More Join: @IITJEE_Advanced

35. Arrange them in decreasing order of their pKa .

(a) 

(b) 

(c) 

(d) 

(A) c > d > a > b (B) b > a > d > c (C) a > b > c > d (D) d > a > c > b

36. During certain drop in energy level from ' n ' to the ground state of H -atom, if frequency of emitted radiation is 0.99 times the frequency of last line of lyman series, then calculate number of Paschen lines if the electron present in (n-2) energy level make transition upto ground level without emitting any Balmer series.

(A) 7 (B) 10 (C) 5 (D) 3

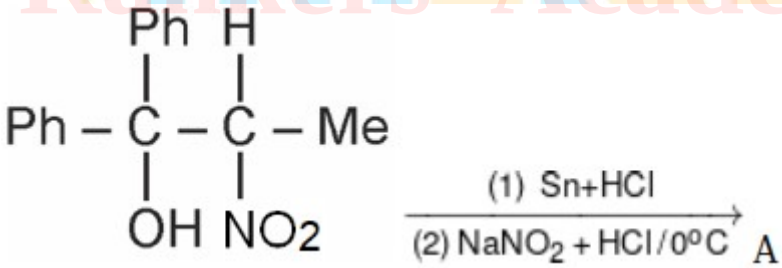
37. Which of the following cation is wrongly matched with its complex color on reaction $K_4[Fe(CN)_6]$ at (assuming no change in oxidation state)

(A) $Co^{+2} \rightarrow$ Green ppt.

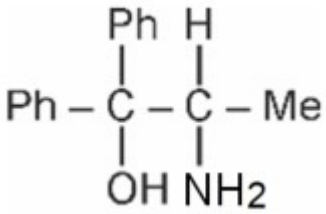
(B) $Mn^{+2} \rightarrow$ White ppt.

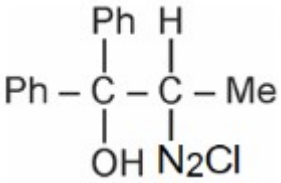
(C) $Fe^{+2} \rightarrow$ Turnbull's Blue ppt.

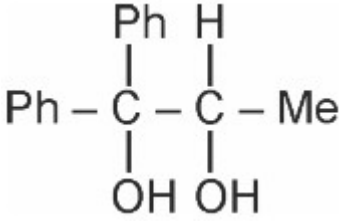
(D) $Cd^{+2} \rightarrow$ Bluish white ppt.

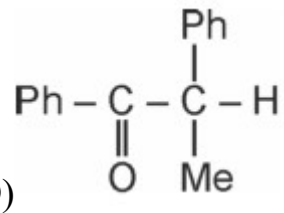
38. 

A is :-

A) 

B) 

C) 

D) 

For More Join: @IITJEE_Advanced

39. An equimolar mixture of $CO \& O_2$ detonated to give carbon dioxide according to the reaction : $2CO + O_2 \rightarrow 2CO_2$

Statement-I : final average molar mass < Initial average molar mass

Statement-II : CO is the limiting reagent

Find correct answer

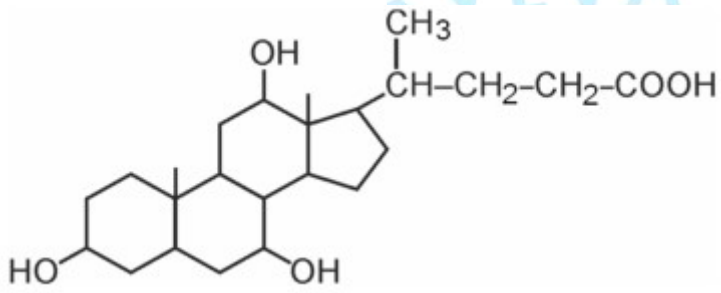
(A) Only statement-I is correct and Statement-II is wrong

(B) Only statement-I and Statement-II are correct

(C) Only statement-I is wrong and Statement-II is correct

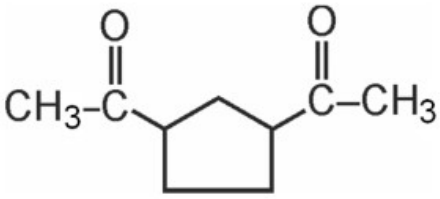
(D) Both are incorrect

40. Calculate total number of stereo isomers of the following



(A) 2^{11} (B) 2^{10} (C) 2^5 (D) 2^6

41. A hydrocarbon C_9H_{14}, santene an important constituent of sandalwood oil on ozonolysis gives compound 'A'.

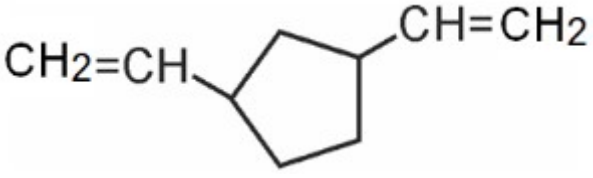


What is the structure of santene?

(A) 

(B) 

(C) 

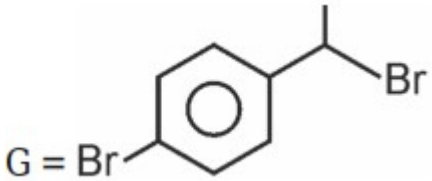
(D) 

42. An aromatic chiral compound $G(C_8H_8Br_2)$ on treatment with aq. NaOH gives $H(C_8H_9BrO)$. On heating G with potassium ter-butoxide $I(C_8H_7Br)$ is formed. With one equivalent of MeMgBr in ether p-bromoisopropyl benzene is formed. Which observation is incorrect about these reactions?

For More Join: @IITJEE_Advanced

(A) I is an optically inactive aromatic alkene

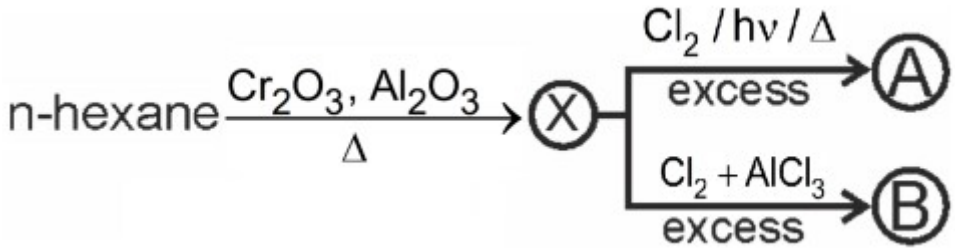
(B) H is an optically active phenol.

(C) 

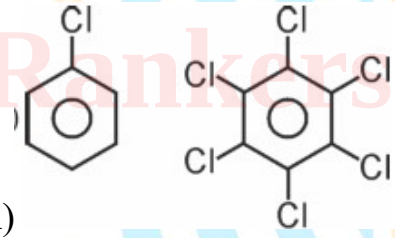
(D) Benzylic halide gives nucleophilic substitution with faster rate than aryl halide.

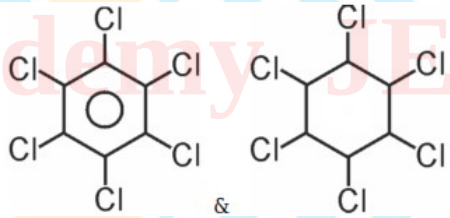
43. Spin only magnetic moment of an Tetrahedral complex of Co^{+2} in the presence of field ligands in B.M. is :

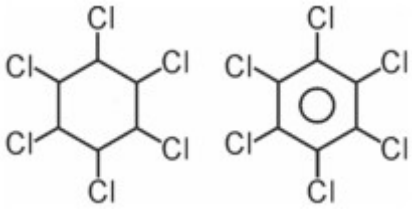
(A) 4.89 (B) 0 (C) 2.82 (D) 3.46

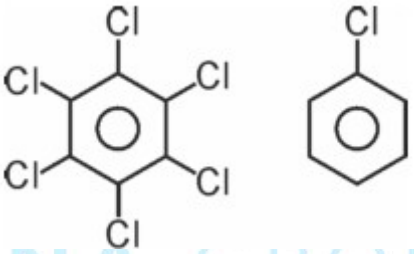
44. 

A & B respectively :-

(A) 

(B) 

(C) 

(D) 

45. Consider following statements & select correct option.

Statement-1: All halogens are coloured.

Statement-2: In halogen absorption of radiations in visible region cause the excitation of outer electron to higher every level.

Statement-3: All halogens absorb different quanta of visible radiation for excitation of electron in high energy level.

(A) Statement-1, Statement-2 & Statement-3 are correct

(B) Only Statement-1 & Statement-2 are correct

(C) Only Statement-2 & Statement-3 are correct

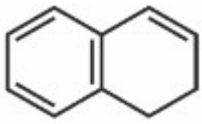
(D) Only Statement-1 & Statement-3 are correct

For More Join: @IITJEE_Advanced

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

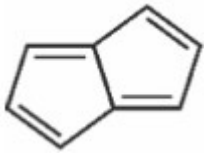
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

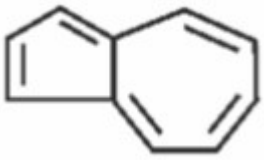
46. How many total number of aromatic compounds are there :-

(1) 

(2) $B_3N_3H_6$

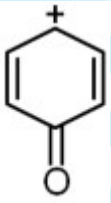
(3) $(C_5H_5)_2Fe$

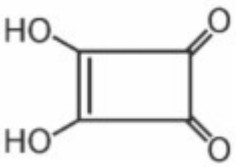
(4) 

(5) 

(6) 

(7) 

(8) 

(9) 

47. For an exothermic reaction :
$2A(g)+B(g)\rightarrow 2C(g)+D(g)$ has an activation enthalpy $30\ kJ\ mol^{-1}$ at 300 K . If energy change ΔE for the reaction is $-20\ kJ\ mol^{-1}$, then the activation enthalpy for the reverse reaction in kJ/mol at 300 K is ____ .
(Given : $R=8\ J\ mol^{-1}\ K^{-1}$)
(Answer to the nearest integer)

48. If the molar concentration of Pb^{2+} ions in aqueous solution prepared by mixing $0.5M, 100ml\ Pb(NO_3)_2(aq)$ solution with $0.5M, 400ml\ KI(aq)$ solution is $(P\times 10^{-O})M$, then the value of Q is_____
(Given : K_{sp} of $PbI_2=8\times 10^{-9}$)

49. Number of the following compound doesn't exist in air or in aqueous solution at normal temperature:-
$FeI_3, CuI_2, VI_2, Co_3O_4, CrO_2, CuF, Mn_2O_7, MnF_6, CrF_6, Cu_2I_2$

50. Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation :
$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$ A current of 10 milli amphere to be passed to 9.65 hr to produced 36.78 mg of potassium chlorate, then calculate the current efficiency in percent.
(Given : Molar mass of $KClO_3=122.6\ g/mol$)

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let ' f' be twice differentiable function defined in R such that $f'(-4)=10$; $f'(4)=-14$ and $f''(x)\geq -3, \forall x\in(-4,4)$.

If $g(x)=\int_0^x f(t)\,dt$ & $f(0)=0$; then

(A) $g(x)$ is increasing and concave down in $(0,4)$

(B) $g(x)$ is decreasing and concave up in $(0,4)$

(C) $g(x)$ is increasing and concave up in $(0,4)$

(D) $g(x)$ is decreasing and concave down in $(0,4)$

52. The value of $\lim\limits_{n\to\infty}\sum_{k=1}^{n}\dfrac{6^k}{\left(3^k-2^k\right)\left(3^{k+1}-2^{k+1}\right)}$ is

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{1}{3}$ (D) 3

53. **Assertion :-** The normal is drawn to ellipse $\frac{x^2}{100}+\frac{y^2}{25}=1$. The maximum radius of circle centered at origin is 5 , when it is touching the normal.

Reason :- Eccentric angle made by point on ellipse is $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

(A) Assertion & reason are true and reason is correct explanation of assertion
(B) Assertion & reason are true & reason is not correct explanation of assertion,
(C) Assertion is true & reason is false.
(D) Assertion is false & reason is true.

54. Let the area of the region $\left\{(x,y):0\leq x\leq 3, 0\leq y\leq \min\left\{x^2+2, 2x+2\right\}\right\}$ be A . Then 12 A is equal to

(A) 160 (B) 164 (C) 163 (D) 170

55. The variance of two data sets A and B are 16 and 25 respectively. The mean of set B is exactly twice the mean of set A . If these two sets are combined into one, with equal number of observation in each set, then the variance of combined data set in terms of μ_A (μ_A the mean of set A) is

(A) $41+\mu_A{}^2$ (B) $\frac{41}{3}+\frac{\mu_A^2}{2}$ (C) $\frac{41}{2}+\frac{\mu_A^2}{4}$ (D) $\frac{41}{4}+\frac{\mu_A^2}{8}$

For More Join: @IITJEE_Advanced

56. O is the centre of the ellipse $\frac{(x-1)^2}{16}+\frac{(y-2)^2}{9}=1$ and P and Q are the two points on the ellipse such that $\angle POQ=90^\circ$, then which of the following is correct.

(A) $\sqrt{OP^2+OQ^2}=25$ (B) $\sqrt{OP^2+OQ^2}=4$

(C) $\sqrt{\frac{1}{OP^2}+\frac{1}{OQ^2}}=\frac{5}{12}$ (D) $\sqrt{\frac{1}{OP^2}+\frac{1}{OQ^2}}=\frac{5}{24}$

57. Tangent is drawn at any point (x_1, y_1) on the parabola $y^2=4ax$. Now tangents are drawn from any point on this tangent to the circle $x^2+y^2=a^2$ such that all the chords of contact pass through a fixed point (x_2, y_2), and $\lambda\left(\frac{x_1}{x_2}\right)+\left(\frac{y_1}{y_2}\right)^2=0$, then value of λ is equal to :-

(A) 1 (B) 2 (C) 3 (D) 4

58. If α,β,γ are roots of $x^3+2x^2-3x+1=0$, then value of $\left[\frac{\alpha\beta}{\alpha+\beta}+\frac{\alpha\gamma}{\alpha+\gamma}+\frac{\beta\gamma}{\beta+\gamma}\right]$ is :(Where [·] denotes greatest integer function)

(A) -1 (B) 0 (C) 1 (D) 2

59. Triangle is formed by joining the vertices of a cube

LIST-I		LIST-II	
(A)	Probability that the triangle is isosceles or equilateral is	(I)	$\frac{2}{7}$
(B)	Probability that the triangle is isosceles but not equilateral is	(II)	$\frac{3}{7}$
(C)	Probability that the triangle is scalene is	(III)	$\frac{4}{7}$
(D)	Probability that the triangle is right angled is greater than or equal to	(IV)	$\frac{6}{7}$

(A) A → II, B → III, C → I, D → I (B) A → II, B → III, C → I, D → IV

(C) A → III, B → II, C → II, D → IV (D) A → II, B → III, C → IV, D → I

60. Let $E=\sum_{k=0}^{1012}\left(\frac{\binom{1012}{k}}{\binom{2024}{k}}\sum_{r=k}^{2024}\binom{r}{k}\binom{2024}{r}\right)$ and R denotes the remainder when E is divided by 49, then R is divisible by ____(where $\binom{n}{r}$ denotes ${}^nC_r$)

(A) 2 (B) 3 (C) 5 (D) 7

For More Join: @IITJEE_Advanced

61. Consider $I_1=\int_0^{\frac{\pi}{4}}(\cot x)e^{-x}dx, I_2=\int_0^{\frac{\pi}{4}}(\cot^2 x)e^{-x}dx, I_3=\int_0^{\frac{\pi}{4}}(\cot^2 x)e^{-x^2}dx$, $I_4=\int_0^{\frac{\pi}{4}}(\cot^3 x)e^{-x^2}dx$; then correct order is :-

(A) $I_1>I_2>I_3>I_4$ (B) $I_4>I_3>I_2>I_1$

(C) $I_4>I_2>I_3>I_1$ (D) $I_4>I_2$ and $I_3>I_1$

62. The number of solutions of the equation $\sin^{-1}(x-1)^2+\cos^{-1}\left(\frac{1}{x^2-1}\right)=\tan^{-1}(x)$ is(are) (Considering only the principal values of inverse trigonometric functions)

(A) 0 (B) 1 (C) 2 (D) 3

63. If normal to the curve $y=x^2$ forms the shortest chord, then the square of slope of normal is equal to :-

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) 2

64. The maximum value of $\left|\frac{\sin^{-1}x+\tan^{-1}x}{\cos^{-1}x+\cot^{-1}x}\right|$ is

(A) $\frac{3}{7}$ (B) 1 (C) 2 (D) 3

65. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\left|\left(\hat{a}+\hat{b}\right)+2\left(\hat{a}\times\hat{b}\right)\right|=2$. If $\theta\in(0,\pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements:

(S1) : $2\left|\hat{a}\times\hat{b}\right|=\left|\hat{a}-\hat{b}\right|$

(S2) : The projection of â on $\left(\hat{a}+\hat{b}\right)$ is $\frac{1}{2}$

(A) Only (S1) is true (B) Only (S2) is true

(C) Both (S1) and (S2) are true (D) Both (S1) and (S2) are false

66. The lines $5x+12y-10=0$ and $5x-12y-40=0$ touch a circle of radius 3 units. If the centre of circle lies in the first quadrant then the coordinates of centre is

(A) $(5,3)$ (B) $(5,1)$ (C) $(5,2)$ (D) $(5,6)$

67. Variable ellipses are drawn with $x=-4$ as a directrix and origin as corresponding focus. The locus of extremities of minor axes of these ellipses is

(A) $y^2=4x$ (B) $y^2=2x$ (C) $y^2=x$ (D) $x^2=4y$

68. Let $f(x)=a\sin x+b(x)^{1/3}+6$, where a and $b\in R$. If $f\left(\log_{30}\left(\log_{15}30\right)\right)=7$, then the value of $f\left(\log_{30}\left(\log_{30}15\right)\right)$ is :

(A) 5 (B) 7 (C) -5 (D) -7

For More Join: @IITJEE_Advanced

69. The number of positive divisors of $\sum_{r=0}^{6}(-1)^{r}{}^{16}C_r$ are

(A) 6 (B) 10 (C) 20 (D) 16

70. Let R be the relation on the set of real numbers such that aRb iff $|a-b|\leq\frac{1}{2}$ then the relation R is

(A) Reflexive and symmetric but not transitive.

(B) Symmetric and transitive but not reflexive.

(C) Transitive but neither reflexive not symmetric.

(D) Equivalence relation.

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Consider the locus of the complex number z in the argand plane given by $\mathrm{Re}(z)-2=|z-7+2i|$. Let $P(z_1)$ and $Q(z_2)$ be two complex number satisfying the given locus and also satisfying $\mathrm{Arg}\left(\frac{Z_1-(2+\alpha i)}{Z_2-(2+\alpha i)}\right)=\frac{\pi}{2}\ (\alpha\in R)$, then the minimum value of PQ is_____

[Where $i=\sqrt{-1}$, and $\mathrm{Re}(z)$ represents real part of complex no. z]

72. Let $y=f(x)$ is a positive real valued function and $\frac{xdy}{y}=\frac{4x^3-y}{3y-2x^3}, f(1)=1$, then $\frac{f(2)}{2}$ equal to_____

73. Let $R\left\{A=\begin{pmatrix}a & b\\ c & a\end{pmatrix}: a,b,c\in\{0,1,2,3,4\}\right\}$ be set of 2×2 matrices. The number of A in R such that A is either symmetric or skew-symmetric or both and $|A|$ is divisible by 5 is :-

74. A bag contains some white and some black balls. All combination being equally likely. Total balls are 10. If three balls are drawn and found to be all black then the probability that the bag contains 1 white and 9 black balls is p , then value of 55 p is equal to____

75. Equation of plane passing through the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ at minimum possible distance from the point $(-1,-3,1)$ is $7x-2y-2z+3=\lambda$, then value of λ is_____

For More Join: @IITJEE_Advanced